

Synopsis

Student Academic Performance Prediction

Project Id: 27

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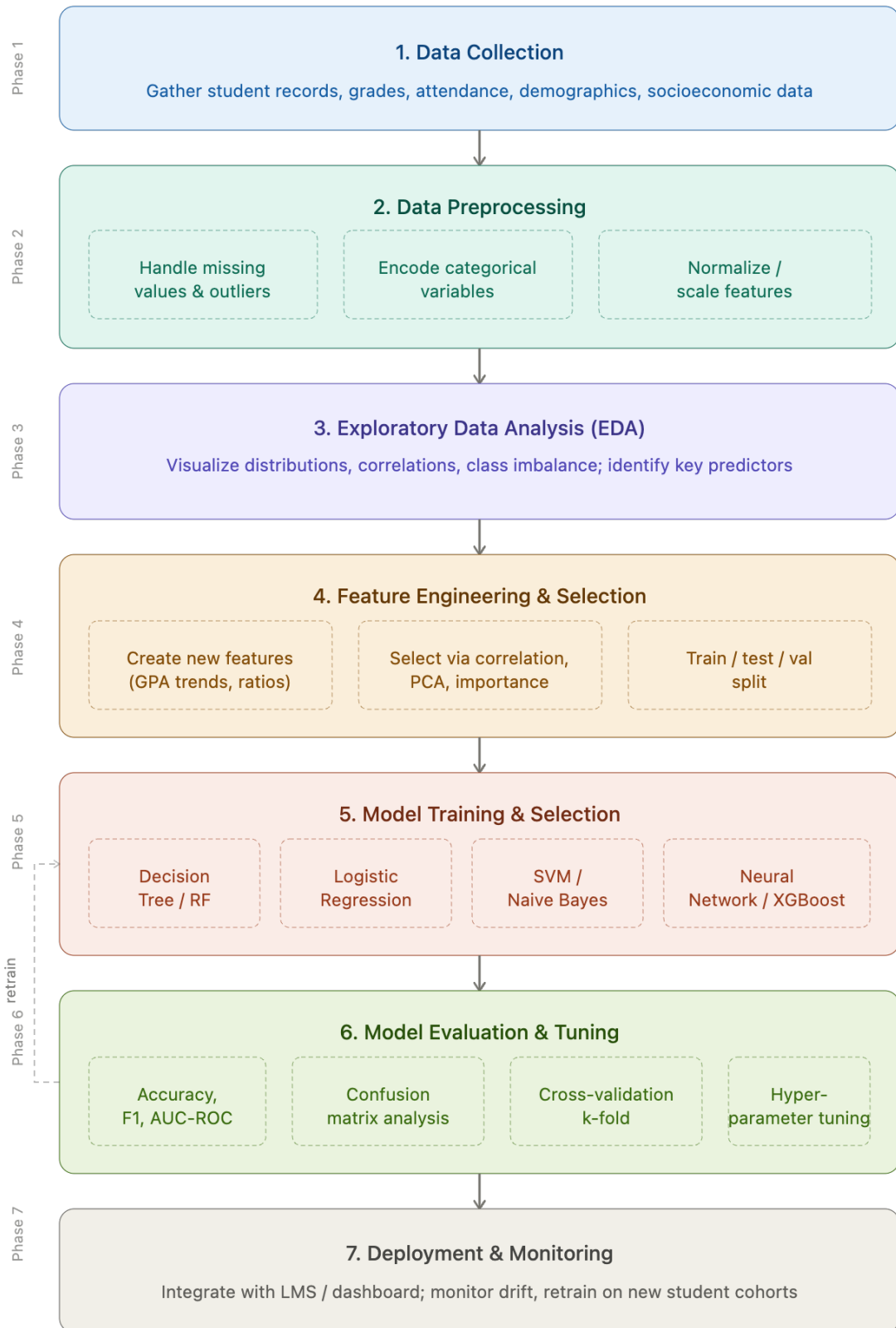
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- **Problem Description:**

Student Academic Performance Prediction is a machine learning task that aims to forecast a student's future academic results (such as grades or pass/fail outcomes) based on past data. This data may include attendance, study time, previous scores, participation, socio-economic background, and other behavioral factors. By analyzing patterns in this data, models like regression or classification algorithms predict how well a student is likely to perform.

In real life, this system helps teachers and institutions identify students who may struggle academically at an early stage. This allows timely intervention such as extra classes, mentoring, or personalized learning plans. It also helps improve overall educational quality by understanding factors affecting performance. Schools and colleges can use these predictions to make data-driven decisions, reduce dropout rates, and support student success. Additionally, it assists parents in tracking progress and encourages students to improve by highlighting areas of weakness.

Steps of Implementation:



- **Proposed Modules:**
 1. **User Authentication Module**
Manages login and access control for students, teachers, and administrators to ensure data security.
 2. **Data Collection Module**
Collects student data such as marks, attendance, assignments, and personal details.
 3. **Data Preprocessing Module**
Cleans and prepares data by handling missing values, removing noise, and converting formats.
 4. **Exploratory Data Analysis (EDA) Module**
Analyzes data using graphs and statistics to identify trends and relationships.
 5. **Feature Selection Module**
Selects the most relevant factors that influence student performance.
 6. **Model Training Module**
Applies machine learning algorithms (e.g., regression, classification) to train prediction models.
 7. **Prediction Module**
Generates predictions about student performance (grades or pass/fail outcomes).
 8. **Model Evaluation Module**
Evaluates model accuracy using metrics like precision, recall, and accuracy.
 9. **Reporting & Visualization Module**
Displays results using charts, dashboards, and reports for easy understanding.
 10. **Feedback & Improvement Module**
Provides suggestions for students and updates the model with new data for better accuracy.

- **Required topics from the subject:**
 1. **Data Preprocessing**
Cleaning and transforming raw data by handling missing values and formatting.
 2. **Exploratory Data Analysis (EDA)**
Understanding data patterns using graphs and basic statistics.
 3. **Feature Selection**
Choosing important variables that affect prediction.
 4. **Classification Algorithms**
Used for categorical outputs like pass/fail (e.g., Logistic Regression).

5. Regression Algorithms

Used for continuous outputs like marks (e.g., Linear Regression).

6. Model Evaluation

Checking performance using accuracy and other metrics.

7. Overfitting & Underfitting

Too complex vs too simple models affecting accuracy.

8. Data Visualization

Using charts/graphs to understand data easily.

- **Platform Required:**

Visual Studio – Development and debugging

Code::Blocks – C/C++ programming

Turbo C++ – Basic academic use

Jupyter Notebook – ML and data analysis

- **Books and link sources:**

Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow

Pattern Recognition and Machine Learning

TensorFlow ML Learning: <https://www.tensorflow.org/resources/learn-ml>

Free ML Books (PDF): <https://infobooks.org/free-pdf-books/computers/machine->